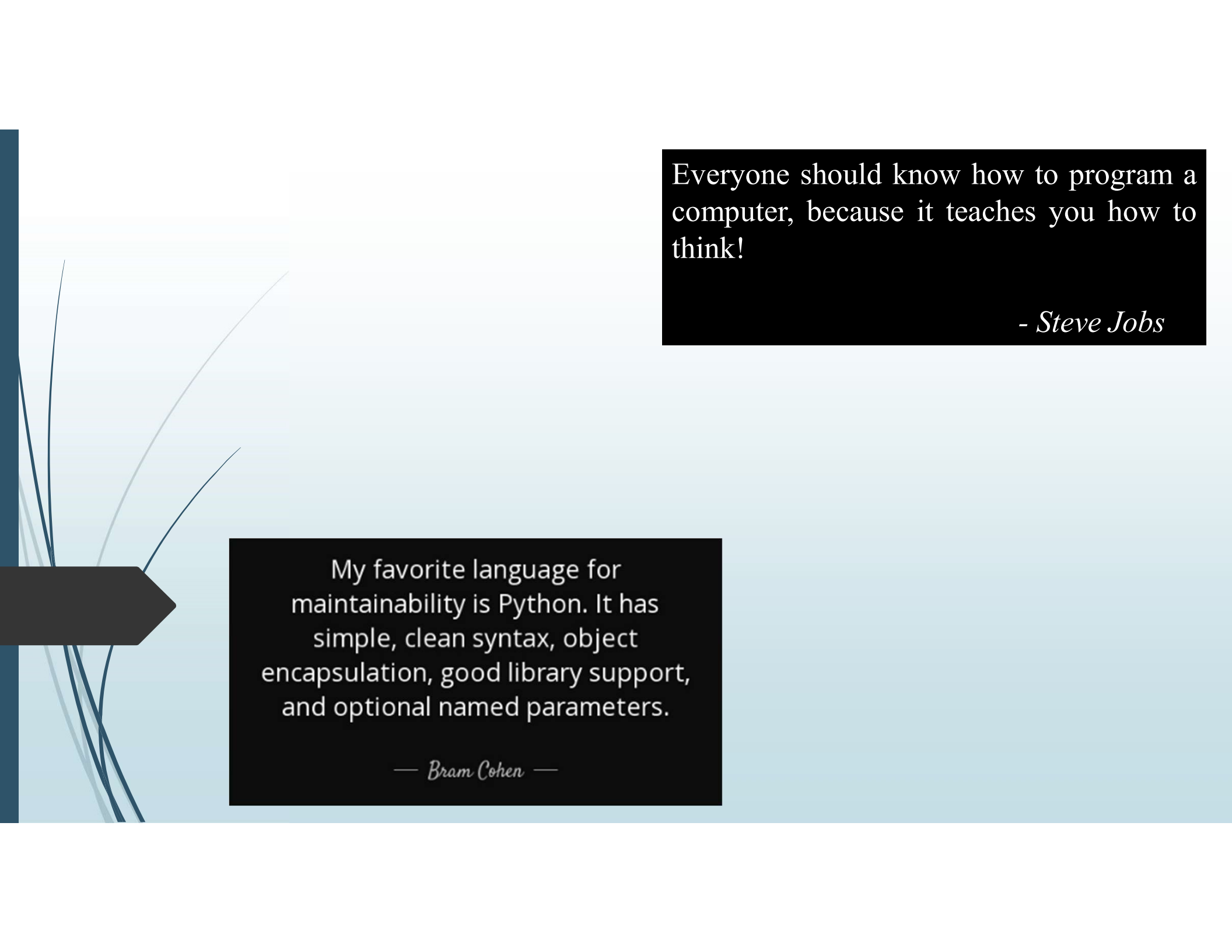




Python

Computer Programming with Python

Harsh Vardhan Nagar
Doctoral Scholar
School of Cyber Security & Digital Forensics
National Forensic Sciences University



Everyone should know how to program a computer, because it teaches you how to think!

- *Steve Jobs*

My favorite language for maintainability is Python. It has simple, clean syntax, object encapsulation, good library support, and optional named parameters.

— *Bram Cohen* —

What is Python?

- 1. High-Level Language:** Python abstracts complex details of the computer from the user, making it easier to write and understand code.



```
Var1 = 0.5  
if var1 > 1.3 or var1 < 0.9:
```

High-level language

Easy for humans to read, write and modify

Programs run slower as they make worse use of the CPU and are not very memory efficient

No understanding of how hardware components work is needed

Examples are python, java, C#, C++ etc.

```
10100101010010101010101  
01010101010010101010100
```

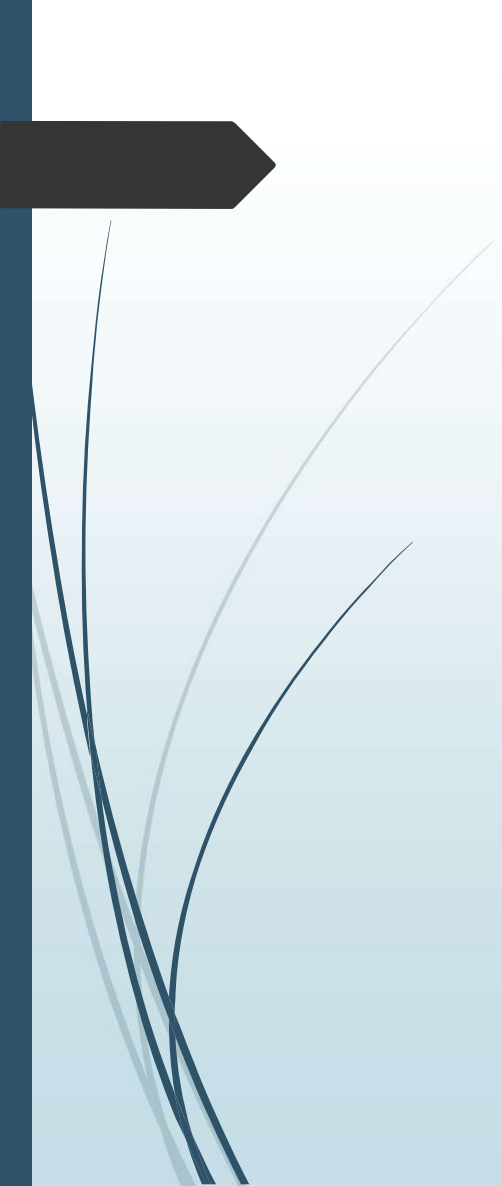
Low-level language

Hard for humans to read, write and modify

Direct control over the CPU means memory use is more efficient and programs run faster

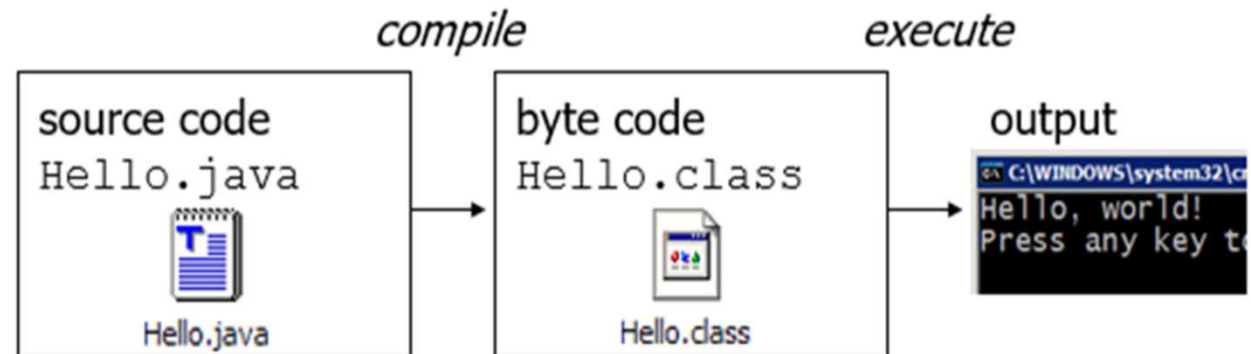
Good understanding of how the different hardware components work is needed

Examples are machine code and assembly language

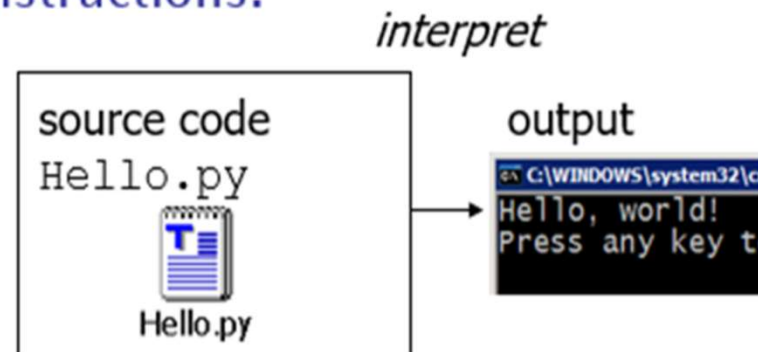


2. Interpreted Language: Python runs code line-by-line through an interpreter, which simplifies debugging and testing.

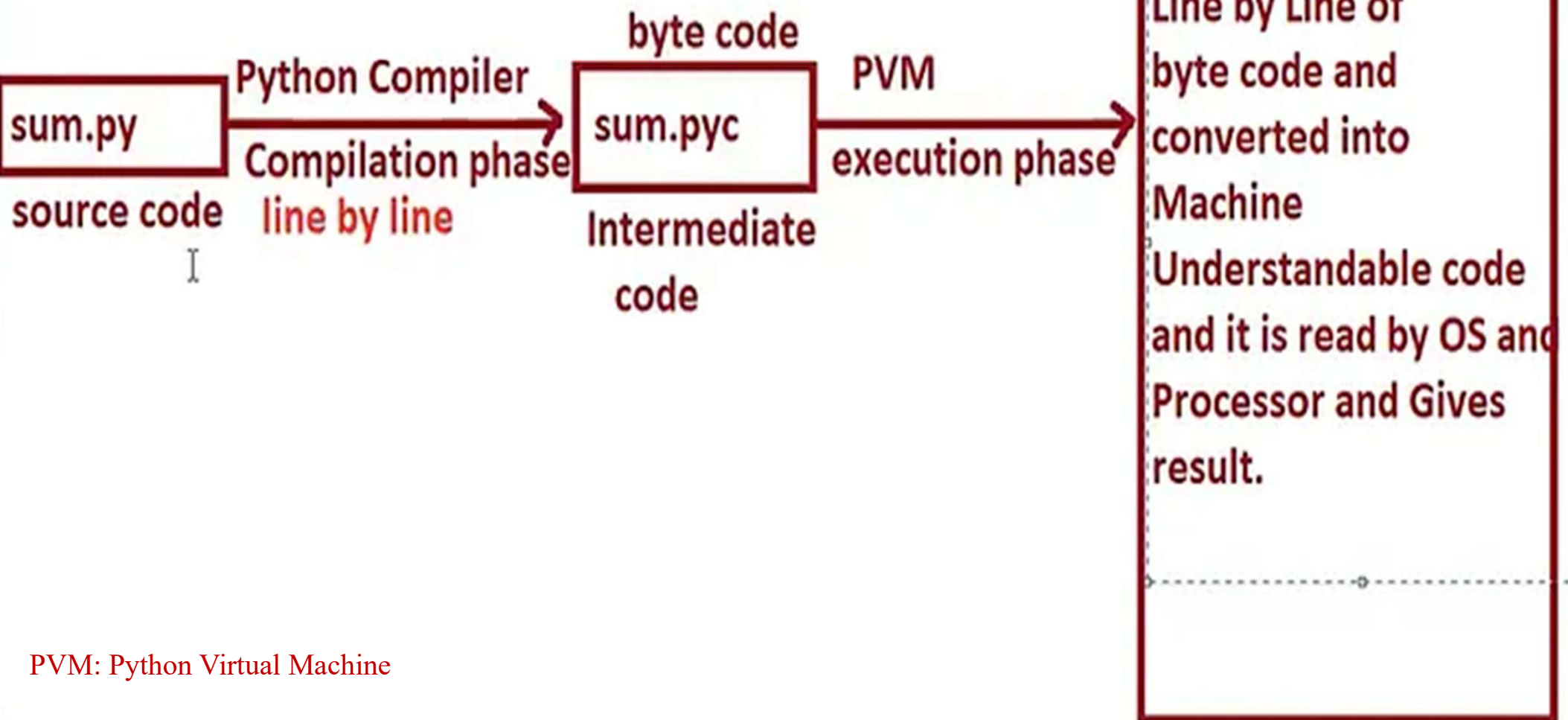
- Many languages require you to *compile* (translate) your program into a form that the machine understands.



- Python is instead directly *interpreted* into machine instructions.



Python Execution Env.



PVM: Python Virtual Machine



3. General-Purpose: It can be used in many domains—web development, data science, AI/ML, automation, scripting, game development, etc.

What is Python used for?





4. Beginner-Friendly Syntax: Python uses clear and readable syntax that resembles plain English, making it flexible and ideal for beginners.

Python VS JAVA Code Comparison

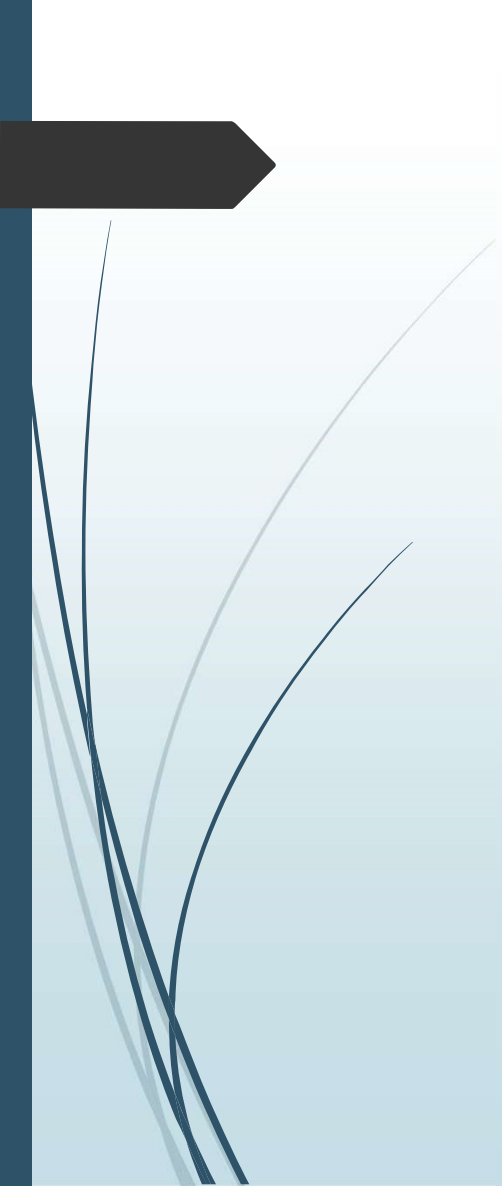
```
name=input("Enter your name: ")
age=int(input("Enter your age: "))
print("Hello," + name + ". You are " + str(age) + " yesrs old.")
```

Python

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter your name: ");
        String name = scanner.nextLine();
        System.out.print("Enter your age: ");
        int age = scanner.nextInt();
        System.out.println("Hello, " + name + ". You are " + age + " years old.");
        scanner.close();
    }
}
```

Java



5. Object-Oriented & Functional:
Supports multiple programming paradigms, including object-oriented, procedural, and functional programming.

Python Programming Paradigms

PROCEDURAL	OBJECT-ORIENTED	FUNCTIONAL
<pre>def area(radius): return pi * radius**2 print(area(5))</pre>	<pre>class Circle: def __init__(self, r): self.r = r def area(self): return self.r**2 c = Circle(5) print(c.area())</pre>	<pre>from math import pi area = lambda r: pi * r**2 print(area(5))</pre>



6. Rich Standard Library: Comes with a vast collection of built-in modules and functions to perform common tasks without extra code

Python

Data Science

- Numpy
- Scipy
- Pandas
- Ipython
- Matplotlib
- PyBrain
- Seaborn
- Scrappy
- Kivy

Networking & digital Forensics

- Socket
- Python nmap
- Pygeoip
- Diesel framework
- Pulsar
- Twisted
- Napalm
- Smtplib
- XMLsax
- Asyncio

Artificial Intelligence

- AIMA
- PyDatalog
- Simple AI
- Easy AI
- NLTK
- gTTS
- Gensim

Machine Learning

- PyML
- MDP-Tool Kit
- Torch
- Accord.NET
- Azure ML studio
- Amazon Machine – Learning
- Scikit-Learn
- PyBrain
- Spark ML Lib

Deep Learning

- Tensorflow
- Theano
- Caffe
- Microsoft CNTK
- Keras
- Pytorch
- Apache-mxnet

IoT

- CoAPthon
- Pyota
- Zorg
- PyFuns
- Paho-mqtt
- mraa



7. Huge Ecosystem: Has thousands of third-party libraries and frameworks (e.g., Pandas, Flask, Scikit-learn) for specialized tasks

Logging

 Loguru
 LogBook
 Structlog
 Picologging

Eye Candy

 Colorama
 Rich

GUI Dev

 PyQt5
 wxPython
 Kivy
 PyForms
 Dear MyGui
 PyAutoGui

Security

 PyCrypto
 pyOpenSSL
 Yara-python
 Scapy

Multimedia

 Pillow
 OpenCV
 PyOpenGL

Data Processing

 Pandas
 Polars
 Datatable
 Modin
 Vaex
 CuPy
 Pydantic

Data Visualization

 Matplotlib
 Seaborn
 Bokeh
 Plotly
 PyGal
 Altair
 Folium

Networking

 Paramiko
 Twisted
 PySocks
 Websockets

Text Process

 TextBlob
 FuzzyWuzzy

Math & Stats

 NumPy
 SciPy
 PyStan
 Pingouin
 Statsmodels
 PyMC3
 CuPy
 Theano

AI & ML

 Scikit-learn
 Tensorflow
 Keras
 PyTorch
 JAX
 XGBoost
 LightGBM
 ELI5
 PyCaret

DB Access

 SQLAlchemy
 PyMongo
 PyMySQL
 Redis-py
 Psycopg2
 HappyBase

Web Development

 Django
 Flask
 Requests
 FastAPI
 Tornado
 Aiohttp
 Pyramid
 Falcon
 CherryPy
 TurboGears
 Web2Py
 Bottle

Distributed Computing

 Dask
 PySpark
 Ray
 Hadoop

Testing

 PyUnit
 Pytest
 Behave
 Lettuce
 Robot
 Testify

Web Scraping

 BeautifulSoup
 Scrapy
 Selenium
 MechanicalSoup
 Tornado

App Dev

 Pretty-errors
 Typer
 Icecream
 BeeWare

IoT Dev

 MicroPython
 CircuitPython
 Bleak
 Paho-mqtt
 Zigpy

Natural Language Processing

 NLTK
 Genism
 spaCy
 BERT
 Polyglot



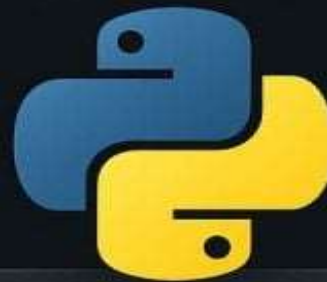
8. Cross-Platform Compatibility:
Python runs on Windows, macOS, and Linux without needing major changes to the code

How to Run a Python Script

macOS

Windows

Linux



```
$ python3 password_generator  
rH3eZ8VV2i
```

```
$ python3 password_generator  
rH3eAZ8VV2i
```

```
C.\>
```

```
$ python password_generator.py  
rH3aAZ8V2i
```





But, Why python...

1. Truly Versatile Language

Python is a multi-purpose language that fits seamlessly into every domain of the modern digital world.

- 
- * **Web Development:** Powering everything from simple blogs to complex enterprise websites using frameworks like Django and Flask.
 - * **Data Science & Machine Learning:** The dominant language for AI and analytics with powerful libraries like Pandas, NumPy, and TensorFlow
 - * **Cybersecurity & Forensics:** Python is used for writing scripts, building penetration testing tools, and automating digital investigations



2. Accessible for All Skill Levels

Gentle learning curve — Simple for beginners, yet deep and powerful for professionals



* Beginner-Friendly Syntax:

Clear, readable, and looks almost like plain English

```
name=input("Enter your name: ")
age=int(input("Enter your age: "))
print("Hello," + name + ". You are " + str(age) + " yrs old.")
```


- * Smooth Transition to Advanced Topics:

As you grow, Python lets you explore complex topics like AI, web development, automation, or cybersecurity without switching languages

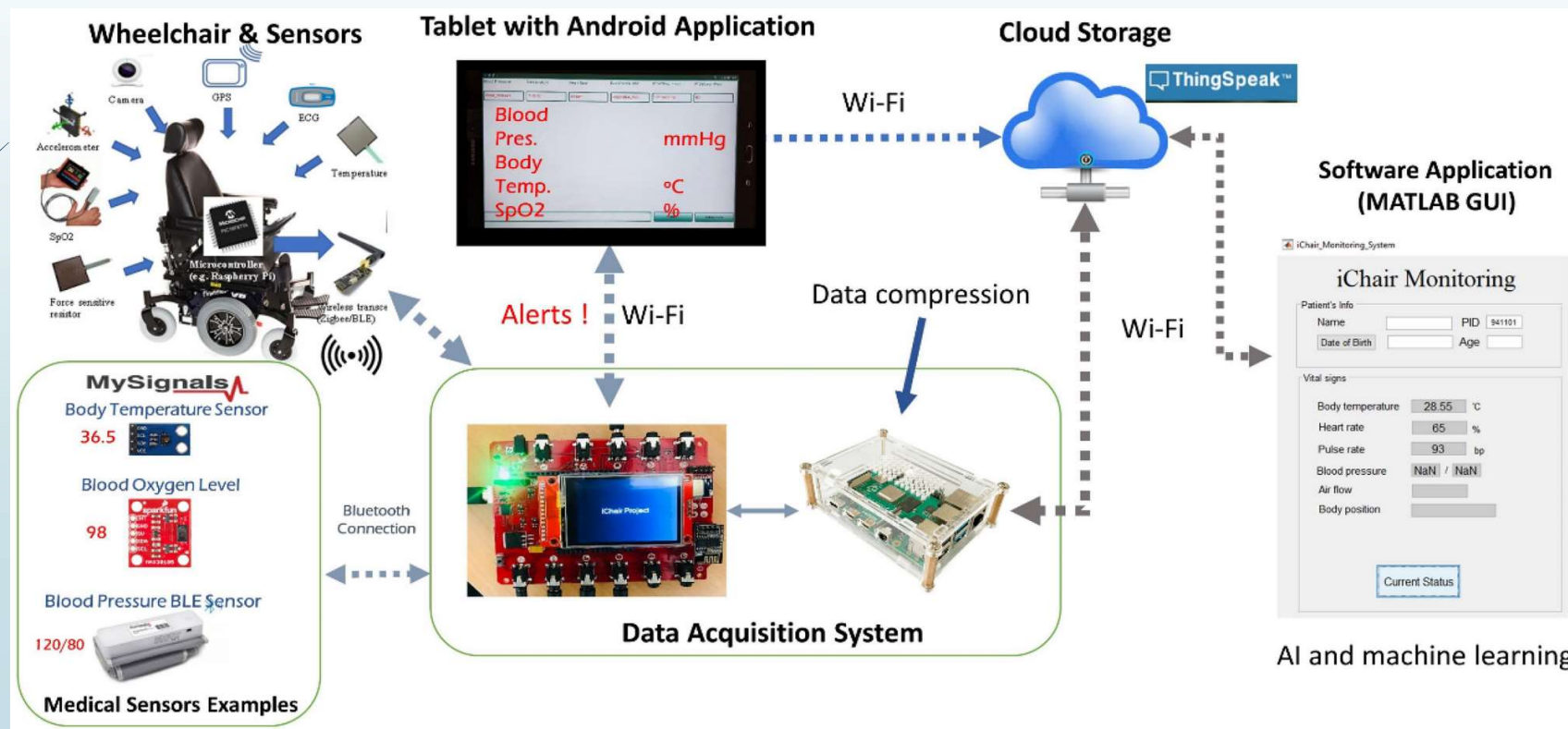


3. Handle Real-World Problems

Packed with production-ready tools that can solve practical, high-impact, real-world challenges.

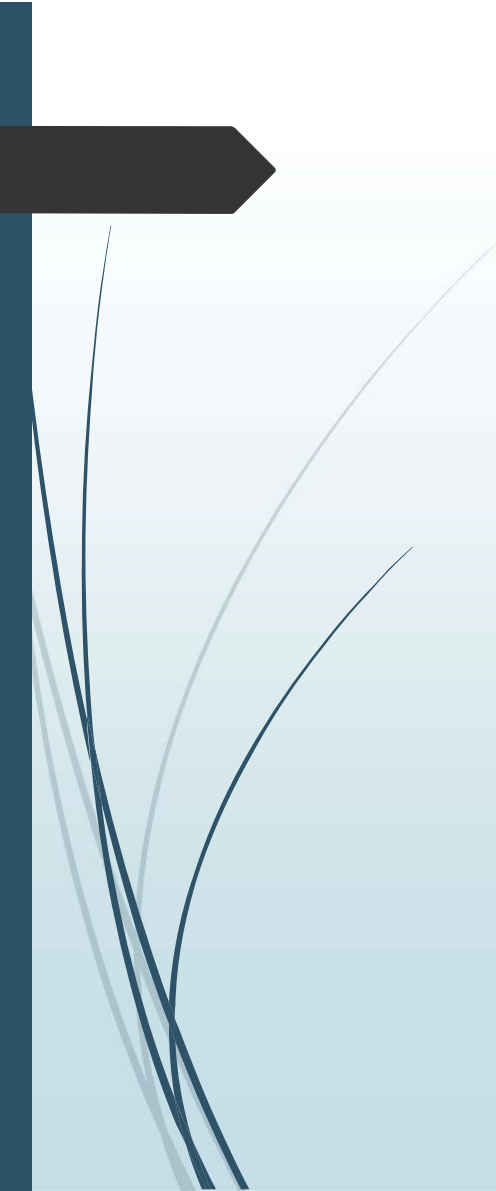
* Personal Health Monitor (Using IoT)

- Tracking key health metrics like weight, temperature, oxygen level, blood pressure.





Can python be used in Cyber
Security & Digital Forensics?



Yes.



1. **Security Tool Development:**

It helps build custom tools for scanning vulnerabilities, malware analysis, penetration testing, and packet sniffing.

2. **Penetration Testing:**

Tools like **Scapy**, **Impacket**, and **Pwntools** are Python-based and widely used for network security assessments.

3. **Automating Evidence Collection:**

Python can automate gathering of logs, memory dumps, disk images, and file system metadata.

4. **Parsing & Analyzing Data:**

Libraries like PyTSK, Volatility, and dfvfs allow forensic experts to analyze hard drives, RAM, and file systems efficiently.

.

PYTHON 2

VS

PYTHON 3

← **Legacy**



Library

0100
0001

ASCII



$7/2=3$



`print "GKTCS"`

Future →

Library



Unicode

0000
0000
0100
0001

$7/2=3.5$



`print ("GKTCS")`

